BoosterPack Ecosystem

Sharp® 128x128 Memory LCD and microSD card BoosterPack
- 1.28 128 x 128 pixel LCD (LS013B7DH03)
- microSD card slot
- DCDC 3V to 5V converter
- Ultra-low-power operation

SimpleLink ULP Sense BoosterPack
- Flow meter measurements
- Ultra low power accelerometer
- Two cap touch buttons
- Analog light sensor
- Reed switch
- 0-200kΩ potentiometer

>>See them all @ ti.com/boosterpack

Software Tools

Professional Software tools
www.ti.com/ccs

LaunchPads are supported by professional IDEs that provide industrial-grade features and full debug-capability. Set breakpoints, watch variables, profile code, inspect memory and more.

Code Composer Studio™ IDE

SmartRF Studio 8
A powerful application to evaluate radio performance.

SimpleLink SDK and SimpleLink Academy
Advanced software and tutorials to easily ramp your development with your new LaunchPad.

© 2023 Texas Instruments Incorporated. The platform bar, LaunchPad, EnergyTrace and Code Composer Studio are trademarks of Texas Instruments.
All other trademarks are the property of their respective owners.

The BoosterPack™ connector

The pinout to connect BoosterPack accessories (available separately) are shown below.

UART (DIO20, DIO22), **Reset** (LP_RST) and **JTAG** (SWCK and SWD) are also present in the LP-EM Debug Connector. Power (**GND**, **3V3** and **5V**) is also provided.
*This function is not connected to the LaunchPad connector by default.

XDS110 EnergyTrace™ Technology

The LP-EM-CC2340R5 is compatible with EnergyTrace technology.

EnergyTrace implements a new method for measuring MCU current consumption. It uses a DC-DC solution to measure the time density of charge pulses, allowing accuracy on ultra low power measurements. Its high dynamic range (700 nA to 400 mA) and fast sampling rate (256 kSPS) captures the complete operational profile of the wireless MCU.

EnergyTrace Profile
EnergyTrace Profile runtime and energy data for low power modes along with each function run during Active Mode.

Graphical Power Data in Code Composer Studio
These two tabs of the EnergyTrace Technology window show a graph over time of power and energy.

Available in the LP-XDS110ET debug probe and selected LaunchPads

Find more information at
ti.com/EnergyTrace

A closer look at your new LaunchPad development kit

Featured microcontroller: CC2340R5

This LaunchPad is great for...

- Battery-operated wireless applications operating in the 2.4GHz ISM RF band
- Add RF capabilities to your product using one of the supported protocols: Bluetooth® 5.3 Low Energy, Zigbee®, SimpleLink™ 15.4 stack and proprietary protocols

What comes in the box?

This Quick Start Guide

10-pin Debug Cable

2-wire Power Cable

CC2340R5 Microcontroller
- 48 MHz CPU
- 512 kB Flash, 36 kB RAM
- Programmable radio supporting various protocols and up to +5 dBm at 2.4 GHz
- Low power consumption
- One 24-bit and three 16-bit timers
- ADC with 12 channels, 12 bits and up to 1.2M samples/s
- Serial communications: UART, SPI, I^2C
- One comparator
- Real-time clock

Hardware setup

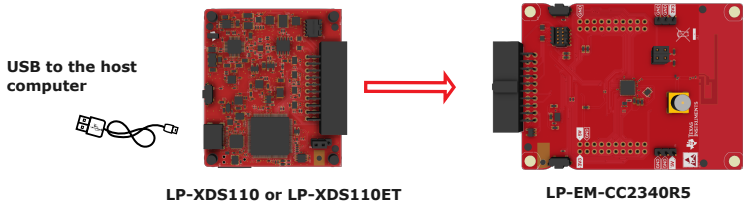
What do you need?

To use your new LaunchPad, you need to connect an external debug probe to either the 20-pin LP-EM debug connector on the edge of the board or to the 10-pin debug connector and supply power separately.

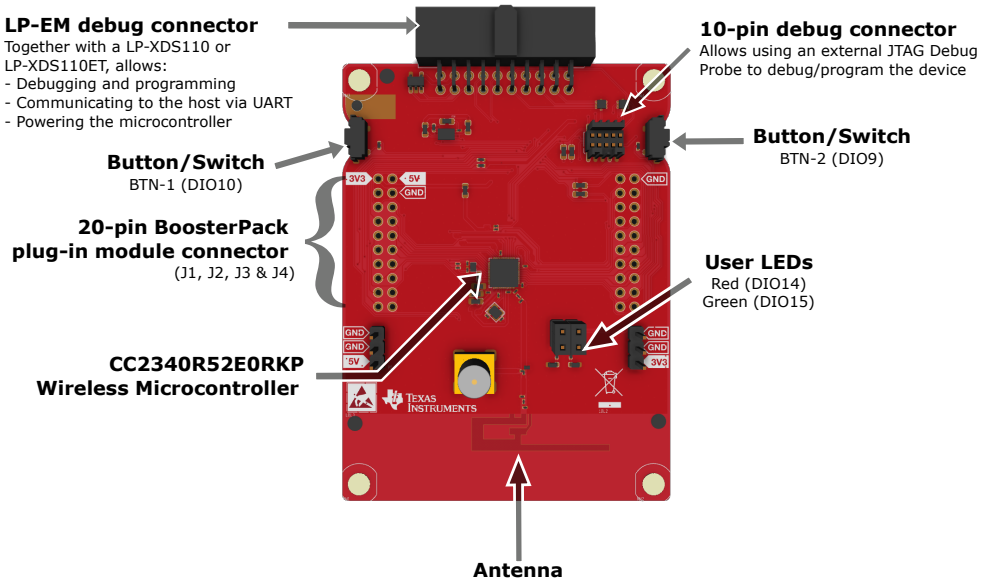
Option 1: Using the LP-EM debug connector

This is the easiest way to setup the hardware. It requires either an **LP-XDS110** or **LP-XDS110ET** debug probe (sold separately).

Simply connect the connector at the edge of the debug probe to its counterpart on the LaunchPad and connect the USB port of the debug probe to the host computer. A secondary UART communications channel will also be available and power to the LaunchPad will be provided directly.

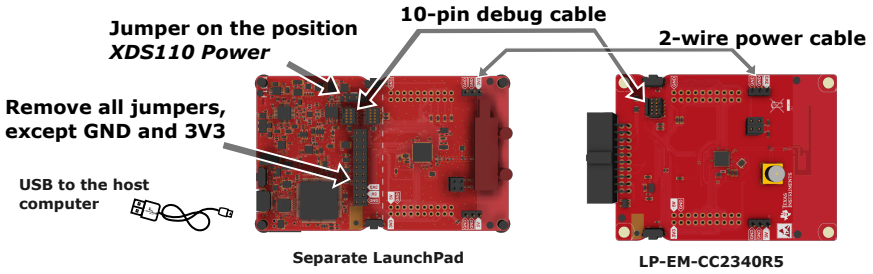


LP-EM-CC2340R5 Overview



Option 2: Using the 10-pin debug connector

Either a standalone debug probe or a separate LaunchPad with a built-in debug probe can be used.
Connect the two boards as shown in the picture below:



For additional details, consult dev.ti.com/?id=LP-EM-CC2340R5

When using the 10-pin debug connector, the UART communications channel must be wired separately (this connector does not carry UART signals).

If using a standalone debug probe, consult its documentation to see if it supports the ARM Cortex-M 10-pin standard.

IMPORTANT NOTICE AND DISCLAIMER

TI PROVIDES TECHNICAL AND RELIABILITY DATA (INCLUDING DATA SHEETS), DESIGN RESOURCES (INCLUDING REFERENCE DESIGNS), APPLICATION OR OTHER DESIGN ADVICE, WEB TOOLS, SAFETY INFORMATION, AND OTHER RESOURCES “AS IS” AND WITH ALL FAULTS, AND DISCLAIMS ALL WARRANTIES, EXPRESS AND IMPLIED, INCLUDING WITHOUT LIMITATION ANY IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE OR NON-INFRINGEMENT OF THIRD PARTY INTELLECTUAL PROPERTY RIGHTS.

These resources are intended for skilled developers designing with TI products. You are solely responsible for (1) selecting the appropriate TI products for your application, (2) designing, validating and testing your application, and (3) ensuring your application meets applicable standards, and any other safety, security, regulatory or other requirements.

These resources are subject to change without notice. TI grants you permission to use these resources only for development of an application that uses the TI products described in the resource. Other reproduction and display of these resources is prohibited. No license is granted to any other TI intellectual property right or to any third party intellectual property right. TI disclaims responsibility for, and you will fully indemnify TI and its representatives against, any claims, damages, costs, losses, and liabilities arising out of your use of these resources.

TI’s products are provided subject to TI’s Terms of Sale or other applicable terms available either on ti.com or provided in conjunction with such TI products. TI’s provision of these resources does not expand or otherwise alter TI’s applicable warranties or warranty disclaimers for TI products.

TI objects to and rejects any additional or different terms you may have proposed.

Mailing Address: Texas Instruments, Post Office Box 655303, Dallas, Texas 75265
Copyright © 2023, Texas Instruments Incorporated